

Determination of Apple, Lemon, and Banana Ripening Stages Using Electronic Nose and Image Processing



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Abstract The objective of this paper is to determine the ripening stages of fruits like the unripe stage, ripeness stage, and overripe stage by using electronic nose and image processing. The electronic nose consists of an array of sensors such as MQ-3, MQ-6, MQ-8, MQ-135 was placed inside the fruit chamber to detect the various odors emitted from the fruits. Each sensor is capable of detecting various gases that were emitted from apple, lemon, and banana. Based upon the value detected by these sensors, ripening stages of these fruit samples were done. Parallel to this image processing technique was done to determine the ripening stages of these fruit samples by their color and shape. Finally, the results obtained by electronic nose and image processing techniques were compared along with manual verification by the human being to improve the accuracy of the ripening stages of apple, lemon, and banana.

Keywords Ripening stages · Electronic nose · Image processing · Sensors · Fruits

1 Introduction

Industrial and agricultural field plays an important role in the economic development of any country. Fruit producers should know the right time for harvesting fruit and at the same time, it should reach the customer before it gets overripe. Fruit's maturity stage is the main factor that affects the fruit quality during ripening and marketability after ripening. Identifying the fruit maturity phase will help the farmers to avoid early harvesting, which means harvesting unripe fruit. It will avoid unwanted transportation of unripe and overripe fruit to save money and time. By proper determination of the fruits, the ripening stage will not allow any wastage of fruit but at the same time, it

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ensures good quality of fruits reaches the market for customers. This might surely help farmers to prevent the unwanted pre-harvest and post-harvest fruits.

2 Literature Review

Prabha [8] Image method technique is to search out the ripening stage like an unripe stage, ripe stage, overripe stage of banana fruit by its color, and size price of their footage precisely. Throughout this paper, the mean color intensity from the bar chart, area, perimeter, major axis length, and minor axis length from the size values was extracted from the activity image. Analysis of variance between each maturity stage on these choices indicated that the mean color intensity and house feature were tons of important in predicting the ripening stage of banana fruit. Their unit of measurement two classifier rules notably mean color intensity recursive rule and house formula were developed and their accuracy on maturity detection was assessed. Hence, this method helps the farmers at the right time of harvest by avoiding pre-harvest of fruit.

Devalatkar and Koli [4] It develops a technique not only to speed up the processing time but also minimizes error in the finding of the age factor of fruits. This proposes an effective fruit grading system based on computer vision techniques. It also determines the defective or non-defective and unripe or ripe or overripe stage of the fruit.

Arakeri [1] It consists of a group of sensors to recognize the odor emitted from fruits. By this, they can eliminate or reduce the lack of quality by using an ethylene gas sensor that can detect fruit maturation. It is also used for continuous monitoring of the freshness of any fruit at the time of sale.

Raut and Bora [9] An electronic nose that is wireless for identifying the ripening stages of fruits. Zigbee, an electronic nose, which is a group of sensors, LabVIEW GUI, and fruits are the sensing system for this project. LabVIEW stand-alone application is for data analysis and principal component analysis-based data classification technique is used to determine the performance of an electronic nose.

Kaur and Gupta [7] It develops a non-destructive instrument that is used for the classification of various fruits. This consists of an array of metal oxide semiconductor gas sensors. The project is based on the response of a sensor array to different VOC emitted from the fruit during its ripening stages. These are analyzed by using PCA and this method is also used for grading and classification of different climacteric fruits, which depends on the ripening stages of fruits.

3 Methodology

The ripening stages of banana, lemon, and apple were determined and classified by using image processing techniques and electronic nose. The result obtained by these two processes was compared to determine the ripening stages. This comparison will

give us a better result. The process to determine the ripening stages is explained below.

3.1 Determining the Ripening Stages Using Image Processing

The objective of this work is to determine the ripening stages (unripe, ripe, over-ripe) of fruits samples of banana, lemon, and apple. For this, the image processing technique is used to classify the numerous ripening stages of fruits samples by mean color intensity. There are several stages involved in the determining of fruit ripeness stages. They are image acquisition, edge detection, RGB value, and classification supported maximum color value. A block diagram of classification for fruit ripening stages is shown in Fig. 1.

3.1.1 Image Acquisition

The acquisition is that the initiative involved the method of classifying the fruits ripening stages. The fruit image is captured by using a camera. This captured fruit image is given as input for this classification process.

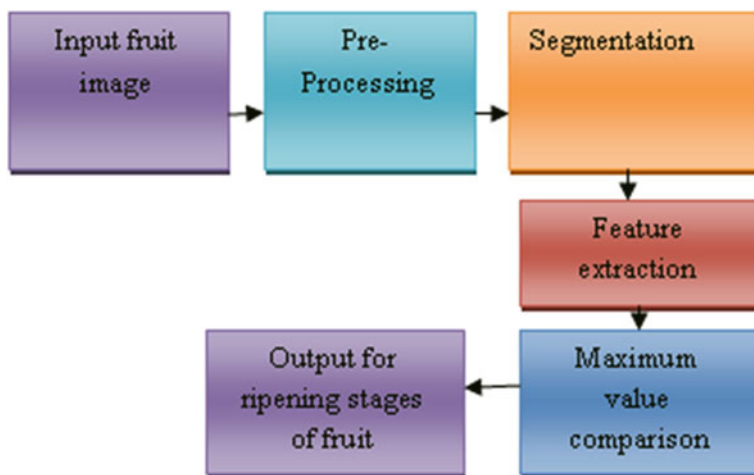


Fig. 1 Block diagram of classification for fruit ripening stages

3.1.2 Edge Detection

Edge detection is one of the image segmentation techniques. This may determine the lines or edges present in an input fruit image. Suthagar et al. [11] The important use of edge detection is to simplify the image data to attenuate the data that are to be processed. There are alternative ways to try edge detection. Most methods are Laplacian edge detection and gradient edge detection. The acceptable sort of edge detection is to be chosen on the idea of requirements for vertical, horizontal tangential edges present within the input image. Sobel filter is employed for this classification work.

3.1.3 RGB Value and Classification

RGB is the format of any color image. The input fruit image is represented with three matrices of sizes matching the image format. Suthagar et al. [10] These matrices correspond to one of the red, green, blue. This RGB value is obtained by using an image processing tool. Then, the maximum color value in an input image can also be obtained. Based upon the maximum color intensity value of an input fruit image, the ripening stages of fruits can be classified.

3.2 *Classification Using Electronic Nose*

This process comprises of electronic nose, fruit chamber, Arduino board, and LCD. After all technical initiation of the electronic nose, the fruit is placed inside the fruit chamber. Geethapriya and Marypraveena [5] Fruit chamber is tightly closed with an array of sensors inside. The sensor array consists of MQ 3 sensor, MQ 6 sensor, MQ 8 sensor, and MQ 135 sensor, which senses the gases emitted from the fruits. Each sensor has the capability of sensing gases in particular ranges. The processor takes the input from the sensors and it displays the value in LCD. The block diagram of classification for fruit ripening stages is shown in Fig. 2.

3.2.1 Array of Sensors

MQ3 sensor is suited for sensing or detecting the gases of ethanol with a standard detection range is 0.1–100 ppm and it has a fast response and high sensitivity. It also has a stable and long life. Kanade and Shaligram [6] MQ6 sensor detects the gases of alkanes (propane, butane), and olefins with a standard detection range are 0.3–50 ppm. It also has a fast response, a stable and long life. MQ8 sensor is suited for detecting the gases of hydrogen with a standard detection range is 0–3 ppm. Chen et al. [3] It also has a stable and long life at low cost. MQ135 sensor detects

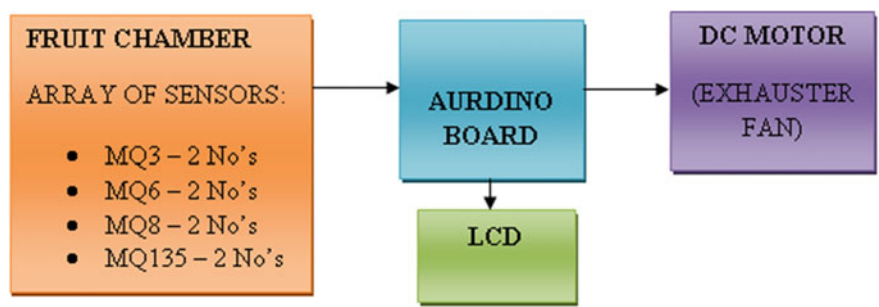


Fig. 2 Electronic nose for classification of fruit ripening stages

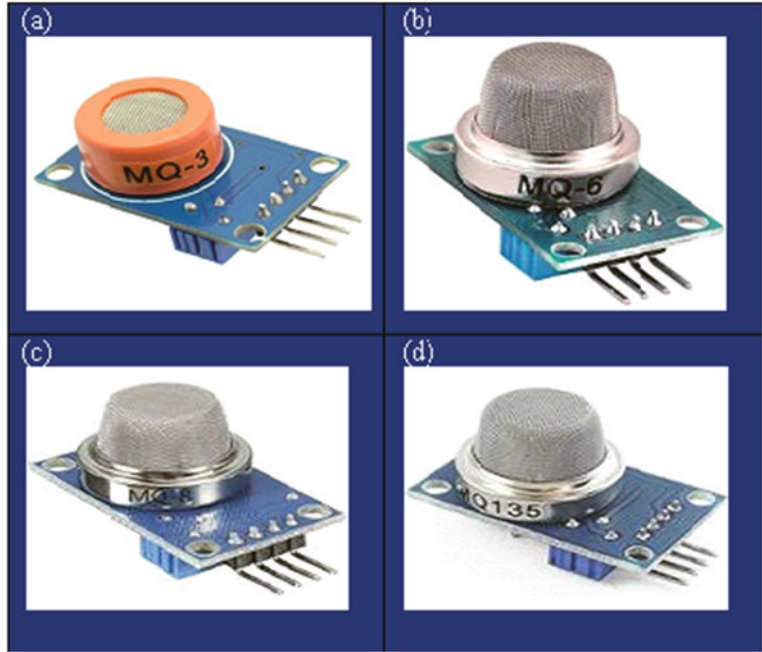


Fig. 3 Shows **a** MQ3, **b** MQ6, **c** MQ8, and **d** MQ135 sensors

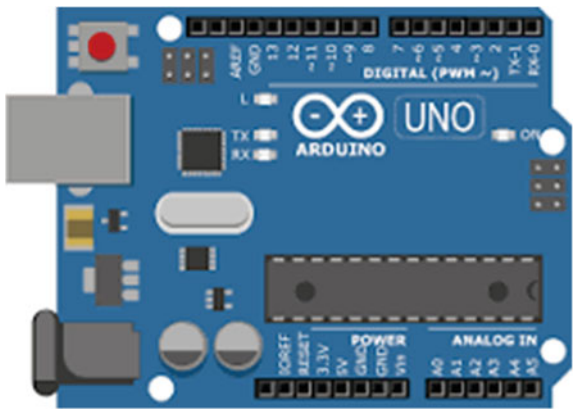
the gases of ammonia, sulfides with standard detection is 0.01–100 ppm. It also has a fast response and high sensitivity stable (Fig. 3).

The target gas, concentration, and operating temperature of MQ3, MQ6, MQ8, and MQ135 sensors are shown in Table 1.

Table 1 Specifications of MQ3, MQ6, MQ8, and MQ135 Sensors

Sensors	Target gases	Concentration	Operating temperature
MQ-3	Alcohol, ethanol, and smoke	(0.1–100) ppm	10 °C–0 °C
MQ-6	LPG, butane, propane, LNG	(0.3–50) ppm	–10 °C–50 °C
MQ-8	Hydrogen	(0–3)ppm	–10 °C–50 °C
MQ-135	NH ₃ , NO _x , alcohol, benzene, smoke, CO ₂	(0.01–100) ppm	–10 °C–45 °C

Fig. 4 Arduino board



3.2.2 Arduino Board

An Arduino board acts as a microcontroller kit. The Arduino pin diagram is shown in Fig. 4. The operation voltage of the Arduino board is 5 V with 14 digital input/output pins and 6 analog input pins. Each digital input pin takes 40 mA and the output pins provide 40 mA DC.

Each analog input/output pin provides 10-bit resolutions. The electronic nose has been interfaced with this Arduino board. Arduino board takes the analog output from the odor sensor and provides the display in LCD.

3.2.3 Operating Procedure of Electronic Nose

The proposed electronic nose system was tested with the odors of three fruits, namely, lemon, banana, and apple. The odors were prepared by placing samples of fruit in a fruit chamber sealed with a membrane. The operational procedures were as follows:

- The fruit chamber was opened and the exhauster fan was ON to prepare the chamber to place fruits. The exhauster fan is ON state to pump the odor present inside the fruit chamber.

- Once the fruit is placed inside the chamber it was closed, and the sensor's resistance was given 60 s to reach a steady state. The classification result obtained by sensors appeared on the LCD.
- The fruit that is placed inside the chamber was removed after taking the reading. After this, empty chamber is kept opened and the exhaustor fan is made to ON state for 60 s.
- After 60 s, another fruit is placed inside the chamber and closed tightly. This process continued for various fruits and reading was noted. The sensors and the exhaustor fan made by DC motor both fit into the fruit chamber and the transparent box, which contains the interface circuitry and the Arduino board.

4 Results and Discussion

4.1 Classification Using Image Processing

The results obtained by the image processing technique which determines the fruit and classifies the ripening stages of various fruits by their shape and color are shown in Fig. 5. Figure 5a, c, and e represents the different ripening stages of banana fruit on various days, and Fig. 5b, d, and f shows the ripening stages of an input banana on various days.

In this system, the ripening stages of various fruits can be classified. Figure 6a,

Fig. 5 a, c, e are the input images and b d, f are the corresponding results

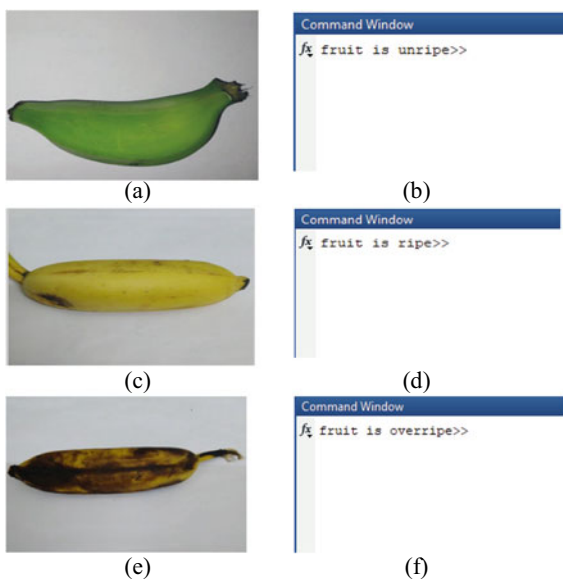
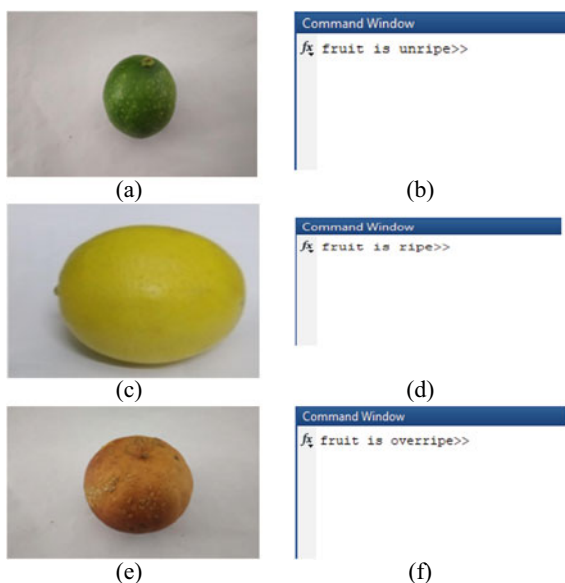


Fig. 6 a, c, e are the input images and b d, f are the corresponding results



c, and e represents the different stages of lemon fruit images, and Fig. 6b, d, and f shows the corresponding ripeness stages of an input lemon fruit, which are given below.

Figure 7a, c, and e represents the different stages of apple fruit images, and Fig. 6b, d, and f shows the corresponding ripeness stages of an input apple fruit, which are given below

4.2 Classification Using Electronic Nose

The ripening stages of various fruits were determined and classified by using an electronic nose. Figure 8 represents the experimental setup of an electronic nose system. Figure 8a, b, and c represents the experimental setup of an electronic nose system for apple lemon and banana fruit.

Unripe banana, lemon, and apple are taken for testing by using an electronic nose, which consists of MQ3, MQ6, MQ8, and MQ135 sensors. Everyday sensor reading is monitored at 6.00 am, 12.00 pm, and 10.00 pm for all three fruits. The average value of each sensor is treated as the day reading of each sensor for that respective fruit. The day-wise and time-wise readings taken from different sensors for three fruits are given in Table 1. It was observed that banana fruit is unripe for the value of the MQ3 sensor lies between 0.0116 ppm and 0.0139 ppm, the MQ6 sensor value lies between 0.6281 ppm and 0.4954 ppm, MQ8 sensor value lies between 0.0060 ppm and 0.0129 and the MQ135 sensor value lies between 0.3288 ppm and 0.3372 ppm. It was also observed that lemon fruit is unripe for the value of MQ3 sensor lies

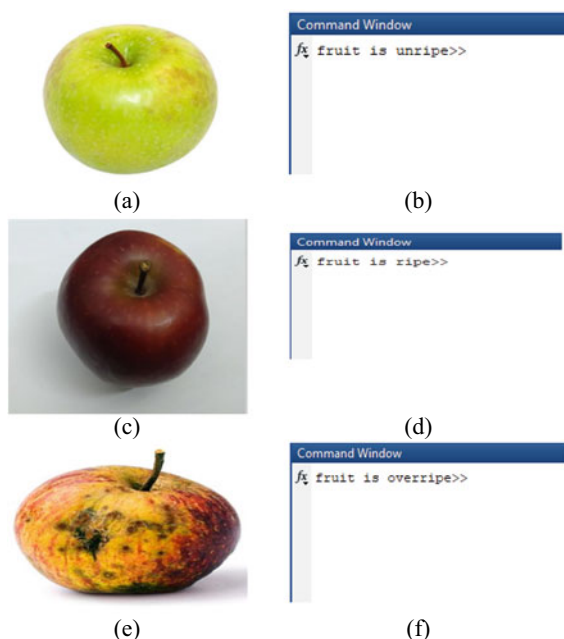


Fig. 7 a, c, e are the input images and b d, f are the corresponding results

between 0.0282 ppm and 0.0307 ppm, MQ6 sensor value lies between 0.5484 ppm and 0.6120 ppm, MQ8 sensor value lies between 0.0060 ppm and 0.0370 and MQ135 sensor value lies between 0.3976 ppm and 0.4550 ppm. Similarly, it was observed that for apple fruit is unripe for the value of MQ3 sensor lies between 0.0044 ppm and 0.0103 ppm, MQ6 sensor value lies between 0.5568 ppm and 0.6205 ppm, MQ8 sensor value lies between 0.0309 ppm and 0.0509 ppm and MQ135 sensor value lies between 0.3962 ppm and 0.4258 ppm.

From Table 2, it was observed that banana fruit is ripe for the value of MQ3 sensor lies between 0.0175 ppm and 0.0253 ppm, MQ6 sensor value lies between 0.0706 ppm and 0.0708 ppm, MQ8 sensor value lies between 0.1599 ppm and 0.1633ppm, and MQ135 sensor value lies between 0.3673 ppm and 0.4037 ppm. It was also observed that lemon fruit is unripe for the value of MQ3 sensor lies between 0.0340 ppm and 0.0415 ppm, MQ6 sensor value lies between 0.7725 ppm and 0.8341 ppm, MQ8 sensor value lies between 0.1641 ppm and 0.1739ppm, and MQ135 sensor value lies between 0.5805 ppm and 0.6565 ppm. Similarly, it was observed that for apple fruit is unripe for the value of MQ3 sensor lies between 0.0338 ppm and 0.0630 ppm, MQ6 sensor value lies between 0.7767 ppm and 0.8016 ppm, MQ8 sensor value lies between 0.1779 ppm and 0.2034 ppm and MQ135 sensor value lies between 0.5825 ppm and 0.6380 ppm.

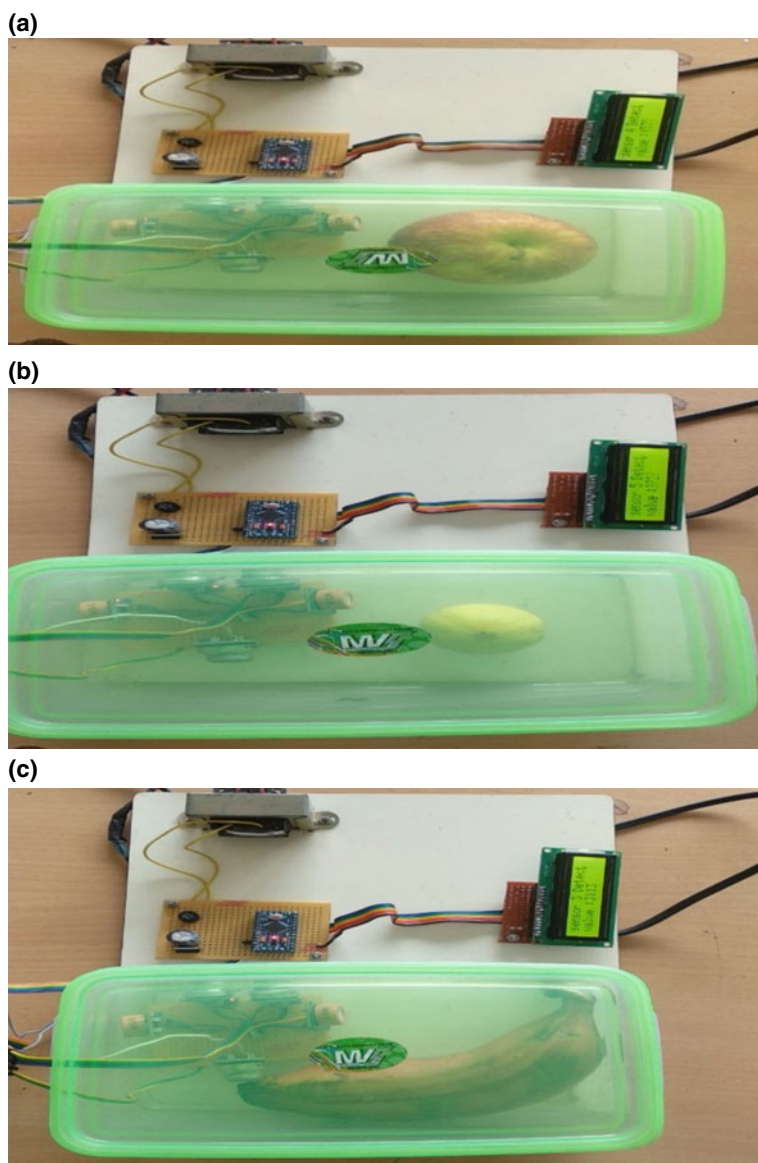


Fig. 8 **a** Electronic nose system for determining the ripening stages of apple fruit **b** Electronic nose system for determining the ripening stages of lemon fruit **c** Electronic nose system for determining the ripening stages of banana fruit

Table 2 Shows the obtained results for different ripening stages of various fruits

Fruits		Banana			Lemon			Apple					
*Sensors		MQ3	MQ6	MQ8	MQ135	MQ3	MQ6	MQ8	MQ135	MQ3	MQ6	MQ8	MQ135
Day 1	6.00am	0.0129	0.6496	0.0034	0.3266	0.0271	0.5466	0.0106	0.3900	0.0012	0.5466	0.0106	0.3900
	12.00 pm	0.0119	0.6345	0.0049	0.3297	0.0277	0.5487	0.0117	0.3922	0.0043	0.5518	0.0223	0.3987
	10.00 pm	0.0101	0.6002	0.0097	0.3301	0.0298	0.5499	0.0229	0.4105	0.0076	0.5721	0.0598	0.3999
	Average	0.0116	0.6281	0.0060	0.3288	0.0282	0.5484	0.0151	0.3976	0.0044	0.5568	0.0309	0.3962
Day2	6.00am	0.0136	0.5991	0.0101	0.3354	0.0301	0.6102	0.0334	0.4376	0.0091	0.5945	0.0421	0.4087
	12.00 pm	0.0137	0.5550	0.0127	0.3367	0.0309	0.6122	0.0376	0.4398	0.0101	0.6212	0.0519	0.4288
	10.00 pm	0.0145	0.3321	0.0159	0.3394	0.0312	0.6137	0.0399	0.4877	0.0117	0.6457	0.0588	0.4399
	Average	0.0139	0.4954	0.0129	0.3372	0.0307	0.6120	0.0370	0.4550	0.0103	0.6205	0.0509	0.4258
Day3	6.00am	0.0154	0.0706	0.1555	0.3652	0.0325	0.7700	0.1603	0.5739	0.0271	0.7654	0.1716	0.5739
	12.00 pm	0.0175	0.0706	0.1568	0.3676	0.0337	0.7721	0.1619	0.5776	0.0287	0.7751	0.1801	0.5834
	10.00 pm	0.0197	0.0707	0.1675	0.3691	0.0358	0.7754	0.1701	0.5899	0.0457	0.7896	0.1819	0.5901
	Average	0.0175	0.0706	0.1599	0.3673	0.0340	0.7725	0.1641	0.5805	0.0338	0.7767	0.1779	0.5825
Day4	6.00am	0.0226	0.0706	0.1616	0.3991	0.0399	0.8101	0.1717	0.6287	0.0567	0.7968	0.1987	0.6197
	12.00 pm	0.0245	0.0706	0.1637	0.4021	0.0407	0.8266	0.1734	0.6534	0.0612	0.8004	0.1998	0.6356
	10.00 pm	0.0287	0.0707	0.1645	0.4099	0.0438	0.8655	0.1765	0.6873	0.0712	0.8076	0.2117	0.6587
	Average	0.0253	0.0708	0.1633	0.4037	0.0415	0.8341	0.1739	0.6565	0.0630	0.8016	0.2034	0.6380
Day5	6.00am	0.0602	0.0127	0.1812	0.4640	0.0588	0.9077	0.1819	0.8131	0.0854	0.9027	0.2603	0.8131
	12.00 pm	0.0634	0.0128	0.1835	0.4677	0.0591	0.9176	0.1843	0.8223	0.0889	0.9077	0.2734	0.8255
	10.00 pm	0.0687	0.0129	0.1899	0.4697	0.0601	0.9366	0.1888	0.8338	0.0912	0.9456	0.2834	0.8459
	Average	0.0641	0.0128	0.1849	0.4671	0.0593	0.9206	0.1850	0.8231	0.0885	0.9187	0.2724	0.8282

(continued)

(continued)

Table 2 (continued)

Fruits		Banana				Lemon				Apple			
*Sensors		MQ3	MQ6	MQ8	MQ135	MQ3	MQ6	MQ8	MQ135	MQ3	MQ6	MQ8	MQ135
Day6	6.00am	0.0717	0.0129	0.1902	0.4802	0.0723	0.9877	0.1904	0.8999	0.0997	0.8967	0.2998	0.8911
	12.00 pm	0.0724	0.0129	0.1937	0.4831	0.0754	0.9901	0.1944	0.9272	0.1003	0.9021	0.3103	0.9110
	10.00 pm	0.0751	0.0129	0.1951	0.4859	0.0793	0.9967	0.1978	0.9577	0.1103	0.9117	0.3288	0.9227
	Average	0.0731	0.0129	0.1930	0.4831	0.0757	0.9915	0.1942	0.9283	0.1034	0.9035	0.3130	0.9083

*All values are in ppm.

As mentioned in Table 2, banana fruit is overripe for the value of MQ3 sensor lies between 0.0641 ppm and 0.0731 ppm, MQ6 sensor value lies between 0.0128 ppm and 0.0129 ppm, MQ8 sensor value lies between 0.1849 ppm and 0.1930 ppm and MQ135 sensor value lies between 0.4671 ppm and 0.4831 ppm. It was also observed that lemon fruit is overripe for the value of MQ3 sensor lies between 0.0593 ppm and 0.0757 ppm, MQ6 sensor value lies between 0.9206 ppm and 0.9915 ppm, MQ8 sensor value lies between 0.1850 ppm and 0.1942 ppm and MQ135 sensor value lies between 0.8231 ppm and 0.9283 ppm. Similarly, it was observed that for apple fruit is overripe for the value of MQ3 sensor lies between 0.0885 ppm and 0.1034, MQ6 sensor value lies between 0.9187 ppm and 0.9035 ppm, MQ8 sensor value lies between 0.2724 ppm and 0.3130 ppm and MQ135 sensor value lies between 0.8282 ppm and 0.9083 ppm.

After measuring the values by various sensors, the ripening stages of all the tree fruits were determined. To improve the efficiency of the system, data collected from the electronic nose are compared with results obtained by image processing technique and manual verification of fruits on the respective days are shown in Table 3. Accuracy of the ripening stages was calculated for this system to show that the ripening stage determination of banana is 94.44%, lemon has 83.33%, and apple has 100%. This method of verifying the fruit ripening stages produces better results when compared with other methods, which were done earlier.

5 Conclusion

This proposed system has better efficiency when compared with other existing systems. The accuracy of ripening stages was well determined by comparing the result produced by electronic nose and image processing techniques. Also to improve the accuracy of the system, it was compared with manual verification by the expert in that field. This system will help farmers to prevent unwanted pre-harvest and post-harvest fruits. This will also reduce the labor work in the packaging phase of the industry. Thus, the determination of apple, lemon, and banana ripening stages using electronic nose and image processing was done with good accuracy.

Table 3 Shows the comparison between the result obtained by the electronic nose, image processing techniques, and manual verification

Fruits	Banana				Lemon				Apple			
	E	IP	M	Accuracy %	E	IP	M	Accuracy %	E	IP	M	Accuracy %
Methods												
Day 1 *Stages	UR	UR	UR	100	UR	UR	UR	100	UR	UR	UR	100
Day 2 *Stages	UR	UR	UR	100	UR	UR	UR	100	UR	UR	UR	100
Day 3 *Stages	R&UR	R&UR	R&UR	100	R&UR	UR	UR	66.66	R&UR	R&UR	R&UR	100
Day 4 *Stages	R&UR	R	R	66.66	R&UR	R&UR	R&UR	100	R&UR	R&UR	R&UR	100
Day 5 *Stages	R	R	R	100	R	R&UR	R&UR	66.66	R	R	R	100
Day 6 *Stages	R	R	R	100	R	R	R	66.66	R	R	R	100

#EN-Electronic Nose; #IP-Image Processing technique; #M-Manual; *UR-Unripe stage; *R&UR-Neither Ripe nor Unripe stage; *R-Ripen stage.

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